15 The escape speed of an oxygen molecule at the Earth's surface is 1.1×10^4 m s^{-1}. What is the escape speed at $4R$ from the centre of the Earth, where R is the radius of the Earth?

A 5.5×10^3 m s^{-1} **B** 6.4×10^3 m s^{-1} **C** 1.1×10^4 m s^{-1} **D** 1.2×10^4 m s^{-1}

Ans: A

At the surface of Earth,

$$\frac{1}{2}mv_{esc}^{\ 2} = \frac{GMm}{R}$$

$$v_{esc} = 1.1 \times 10^4 = \sqrt{\frac{2GM}{R}}$$

At $4R$ from Earth's centre,

$$\frac{1}{2}mv_{esc}'^{2} = \frac{GMm}{4R}$$

$$v_{esc}' = \sqrt{\frac{2GM}{4R}} = 1.1 \times 10^4 \times \frac{1}{2} = 5.5 \times 10^3 \text{ m s}^{-1}$$